
EduGenie

Google Gemini Powered Learning Assistant

Phase 1: Brainstorming & Ideation

1. Introduction

EduGenie is a lightweight AI-powered educational assistant designed to simplify learning through generative AI. The project is intended for students and learners who need quick answers, simplified explanations, quizzes, summaries, and structured learning recommendations in one place.

The project combines a FastAPI backend with a simple web interface. Gemini-based cloud AI is used for tasks such as question answering, summarization, quiz generation, and learning-path recommendations, while a local lightweight model is used for concept explanation. This hybrid approach ensures both quality and accessibility for learners across different educational levels.

2. Problem Identification

Students often face the challenge of using different resources for different learning activities. A typical learning workflow requires a learner to use one resource for answering academic questions, another for understanding difficult concepts, yet another for creating practice questions, and additional resources for summarizing material or planning what to study next. This fragmented approach to learning creates inefficiencies and adds unnecessary complexity to the study process.

The project identifies the critical need for a **single, unified learning assistant** that can provide multiple learning-support capabilities through an accessible web interface. By consolidating these functions, students can streamline their learning process and spend more time on actual learning rather than navigating between different tools and platforms.

3. Brainstorming & Idea Development

Through systematic analysis of student learning needs, the following brainstorming process identified specific problems and corresponding AI-powered solutions:

Identified Need	Brainstormed Solution
Quick answers to academic questions	Provide a question-and-answer feature powered by generative AI
Difficulty understanding complex concepts	Provide simplified explanations for learning concepts
Need for self-practice	Generate quizzes with multiple-choice questions and answer checking
Large passages difficult to review quickly	Provide a summarization feature for educational content
Learners may not know what to study next	Generate structured learning recommendations and stepwise learning paths

4. Proposed Solution

EduGenie is proposed as a comprehensive web-based AI learning assistant that brings multiple learning-support functions into a single, intuitive interface. The system eliminates the need for learners to switch between different tools and platforms.

The user workflow is straightforward and accessible: the learner selects a learning task from the available options, enters the required text or question, and submits the request to receive an AI-generated result. This simple workflow ensures that learners of different technical backgrounds can easily use the system without complex procedures or configurations.

EduGenie integrates five primary learning-support functions, each powered by AI and designed to address specific learning needs:

- **Explain** – Simplifies complex concepts into easy-to-understand explanations
- **QnA** – Provides quick and accurate answers to academic questions
- **Quiz** – Generates practice quizzes for self-assessment and learning reinforcement
- **Summary** – Creates concise summaries of lengthy educational content
- **Recommend Path** – Provides structured, stepwise learning guidance and resource recommendations

5. Target Users

EduGenie is designed for a diverse range of learners:

- **Students at different academic levels** who need quick learning support during their studies, from secondary education through higher education
- **Self-learners and independent learners** who want structured guidance and practice without a formal educational setting
- **Learners struggling with complex concepts** who need difficult material explained in simpler, more accessible language
- **Active students** who need summaries, quizzes, and learning recommendations while studying specific subjects or preparing for examinations

The platform is designed to be accessible to learners regardless of their technical background or prior experience with AI tools.

6. Project Objectives

The primary objectives of the EduGenie project are clearly defined:

- Create a simple, accessible AI-powered educational assistant for comprehensive learning support
- Provide concise and useful answers to student questions with minimal response time
- Simplify complex concepts so they are easier to understand for learners of different levels
- Generate quizzes automatically for self-practice and answer verification
- Summarize educational passages and study materials to make revision and review more efficient
- Provide structured learning recommendations and stepwise learning guidance for informed study planning
- Combine these capabilities into a lightweight and accessible web application that requires no installation or technical setup

7. Key Features Identified During Ideation

Feature	Purpose
QnA	Answer academic questions using generative AI with accurate and contextual responses
Explain	Provide simplified explanations of complex concepts to improve understanding
Quiz	Generate three multiple-choice questions with four options each and provide answer checking functionality
Summary	Summarize longer educational passages and study materials for quick review
Recommend Path	Provide resources, structured learning paths, and stepwise guidance for organized learning
Web Interface	Provide a simple task-selection, input, submission, and result-display workflow accessible from any browser

8. Initial Project Vision

The initial vision for EduGenie is to provide a **lightweight, accessible, and practical learning assistant** that helps users move efficiently from a question or study material to an actionable learning response. Rather than requiring learners to navigate multiple platforms and tools, EduGenie consolidates essential learning functions into one cohesive system.

The project focuses on combining AI-powered assistance with a simple, intuitive interface so learners can use the system without complicated workflows or technical knowledge. The emphasis is on practicality and usability, ensuring that the tool enhances the learning experience rather than adding complexity to it.

By leveraging modern AI technologies such as Google Gemini and FastAPI, EduGenie aims to demonstrate how generative AI can be responsibly applied to educational support while maintaining simplicity and accessibility as core design principles.

9. Expected Outcome

Upon successful completion of all project phases, EduGenie is expected to deliver the following outcomes:

- A **functional web-based learning assistant** with all identified learning-support features implemented and tested
- A **simple, intuitive interface** through which users can select a task and receive results without technical complications
- **AI-generated answers, explanations, quizzes, summaries, and learning recommendations** that provide genuine educational value
- A **stable foundation** that can be extended with additional educational capabilities such as progress tracking, personalized learning paths, and multi-language support
- Complete project documentation suitable for college submission, GitHub repository hosting, and future reference

10. Conclusion

The brainstorming and ideation phase has successfully established **EduGenie as a unified, AI-powered learning assistant** aimed at reducing the effort required to access different types of learning support. By addressing specific student needs identified through systematic analysis, the project demonstrates clear value and practical application.

The identified problems, proposed solution, target users, project objectives, and initial features provide a solid foundation for the following project phases:

- Requirement Analysis – Detailed specification of system requirements and user expectations
- Project Design – Architecture design and technical implementation planning
- Project Planning – Timeline, resource allocation, and risk management
- Project Development – Coding, integration, and feature implementation
- Project Testing – Quality assurance, bug fixing, and performance optimization
- Project Documentation – Comprehensive technical and user documentation
- Project Demonstration – Live demonstration and project presentation

This Phase 1 document serves as the baseline for all subsequent project phases and provides the strategic direction for EduGenie development.

